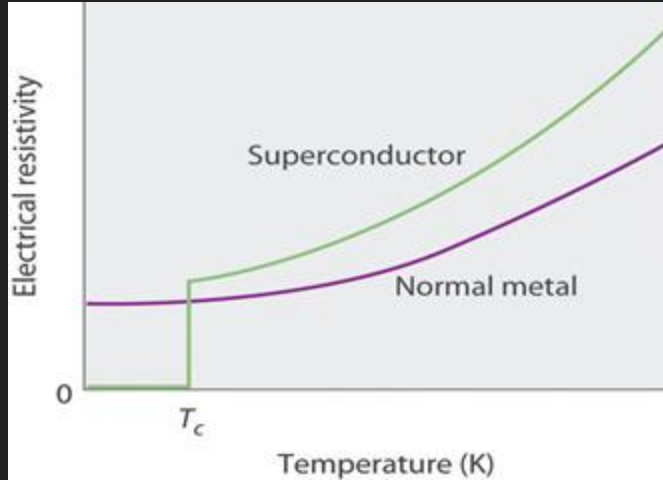


Machine learning modeling for high critical temperature superconductor discovery

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<https://github.com/drooryck/superconductors-ml>

Motivation: Superconductors!



What is a superconductor?

- Exhibits no electrical resistivity under a critical temperature (T_c)
- Current known materials must be cooled to 9K (-260°C)

What are applications?

- High-speed rail
- MRI machines
- Microchips

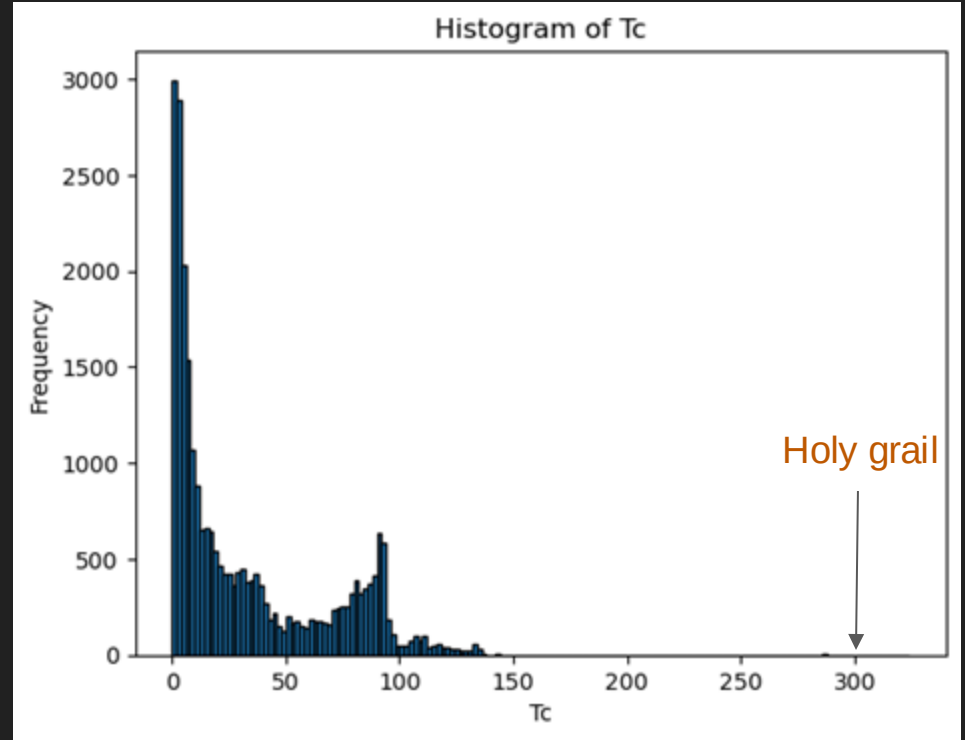


Approach: Computational Materials Science

Databases: SuperCon

- Most extensive of its kind (26.000 entries)
- Records T_c for experimentally studied materials
- Many low- T_c , some high- T_c materials

MDR MATERIALS
DATA
REPOSITORY



Databases: SuperCon

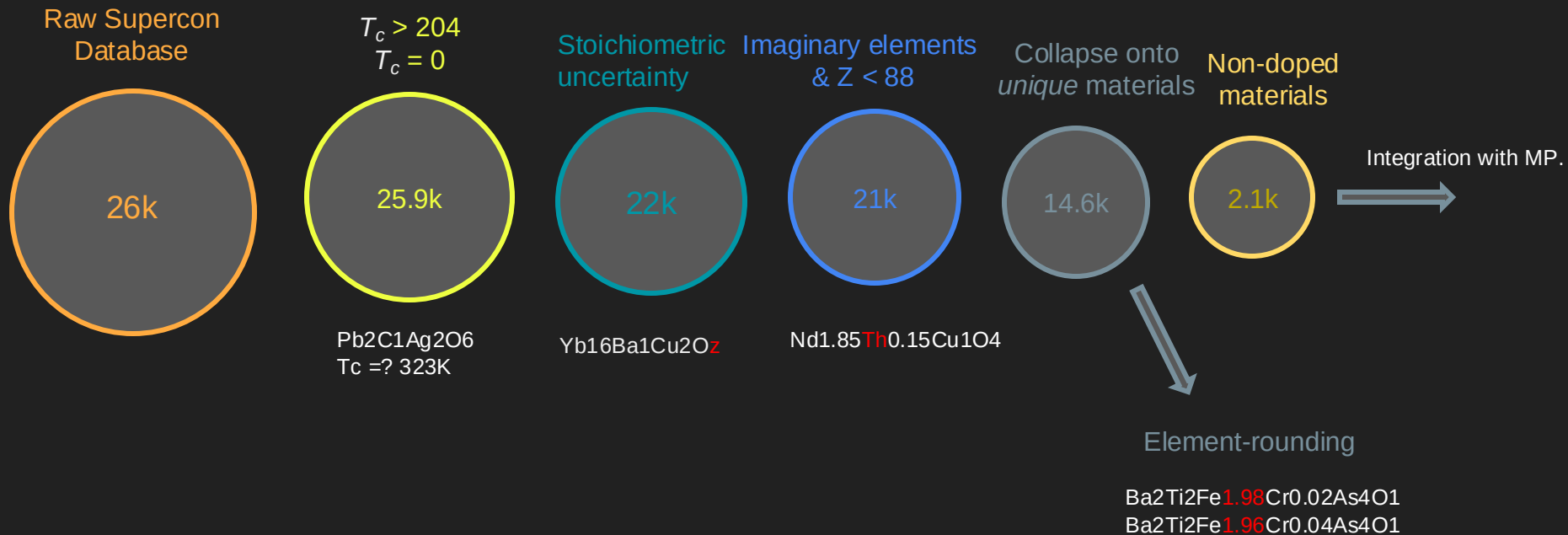
data number	common formula	chemical formula	common name of structure	unit of Tc	Tc	journal
1	2	(Ba,La) ₂ CuO ₄	Ba _{0.2} La _{1.8} Cu ₁ O ₄ -Y	T	K 29.0	Jpn.J.Appl.Phys., 26(1987)L223
2	3	(Ba,La,Ag) ₂ CuO ₄	Ba _{0.1} La _{1.9} Ag _{0.1} Cu _{0.9} O ₄ -Y	T	K 26.0	Jpn.J.Appl.Phys., 26(1987)L223
3	4	(Ba,La) ₂ CuO ₄	Ba _{0.1} La _{1.9} Cu ₁ O ₄ -Y	T	K 19.0	Jpn.J.Appl.Phys., 26(1987)L223

26.000 entries

Main drawbacks:

- Bad data!
 - Troves of missing values
 - T_c not recorded under same conditions
 - Human error in recording chemical formula
 - No descriptors

Databases: Pre-processing pipeline



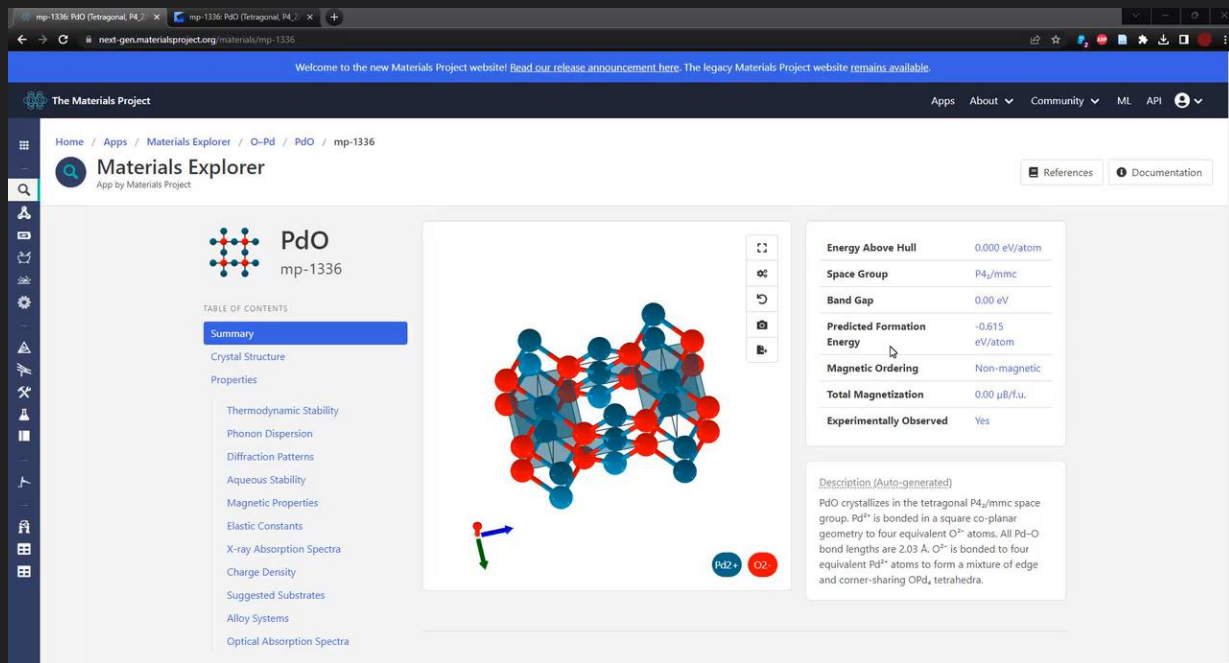
Databases: Matching materials with vast repositories

Density-Functional Theory Databases

<https://next-gen.materialsproject.org/>

Highlight: Materials Project:

- 150.000+ structures
- Lots of useful descriptors
- 800 matches with SuperCon



Welcome to the new Materials Project website! [Read our release announcement here.](#) The legacy Materials Project website remains available.

The Materials Project

Home / Apps / Materials Explorer / O-Pd / PdO / mp-1336

Materials Explorer
App by Materials Project

References Documentation

PdO
mp-1336

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3D Crystal Structure Model

Energy Above Hull	0.000 eV/atom
Space Group	P4 ₂ /mmc
Band Gap	0.00 eV
Predicted Formation Energy	-0.615 eV/atom
Magnetic Ordering	Non-magnetic
Total Magnetization	0.00 μ_B /f.u.
Experimentally Observed	Yes

Description (Auto-generated)

PdO crystallizes in the tetragonal P4₂/mmc space group. Pd²⁺ is bonded in a square co-planar geometry to four equivalent O²⁻ atoms. All Pd-O bond lengths are 2.03 Å. O²⁻ is bonded to four equivalent Pd²⁺ atoms to form a mixture of edge and corner-sharing OPd₄ tetrahedra.

Machine learning: multi-stage pipeline

Does XYZ superconduct at all?

If so...

At what temperature does XYZ superconduct?

76.000 novel materials



Room-temp
superconductor!

Model A:

- Classification
- Trains on 14k superconductors & 76k insulators
- Out-of-sample 94% accuracy



Model B:

- Regression on T_c
- Trains on T_c of 14k superconductors
- Out-of-sample *RMSE* of $\pm 9.5\text{K}$, $R^2 = 0.93$

Machine learning: 1) compositional descriptors

Features are **statistics of the properties of constituent elements**

Atomic Weight
First Ionization Energy
Covalent Radius
Density
Electron Affinity
Enthalpy of Fusion
Specific Heat Capacity
Number of Valence Electrons

Table: Elemental properties used in compositional models

Example: **MgB₂**



	Magnesium	Boron
# element	1	2
# valence electrons	2	3



Feature: Weighted average of # Valence e- : **2.66**

What gives the model predictive power?

Tried different models:

- Gradient-boosted decision trees work best
- Decently interpretable, scalable, and fast

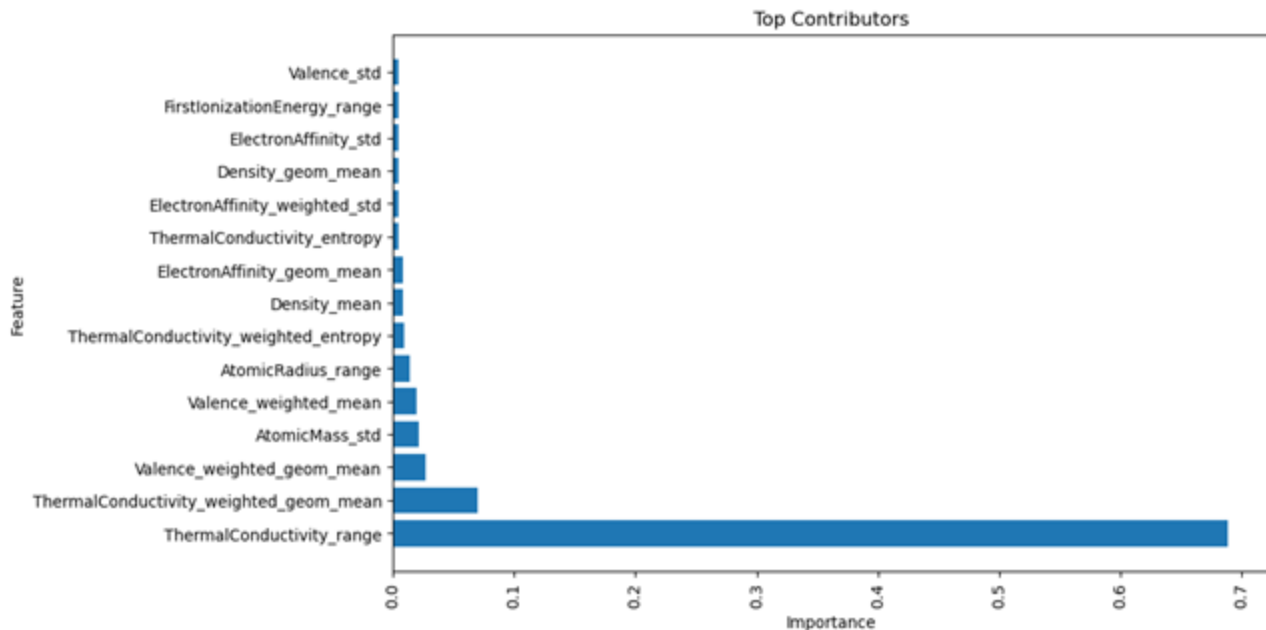
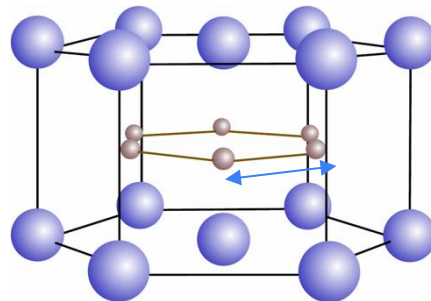
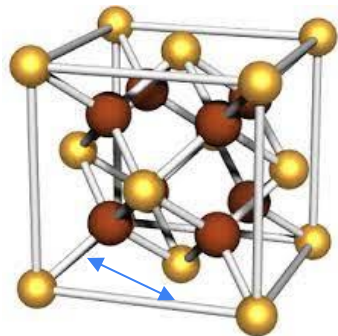


Fig: Feature-importances in regression model to predict Tc

Machine learning: 2) structural descriptors



Featurization from **unit cell of the crystal**:

- Vector of Minimum Relative Bond Distances
- State of the art: **Crystal-Graph Convolutional Neural Networks**, Xie and Grossman (2018)
- Limited by only having 800 matches so far

Machine learning: some previous results

Table 3. List of potential superconductors identified by the pipeline

Compound	ICSD	SYM
CsBe(AsO ₄)	074027	Orthorhombic
RbAsO ₂	413150	Orthorhombic
KSbO ₂	411214	Monoclinic
RbSbO ₂	411216	Monoclinic
CsSbO ₂	059329	Monoclinic
AgCrO ₂	004149/025624	Hexagonal
K _{0.8} (Li _{0.2} Sn _{0.76})O ₂	262638	Hexagonal
Cs(MoZn)(O ₃ F ₃)	018082	Cubic
Na ₃ Cd ₂ (IrO ₆)	404507	Monoclinic
Sr ₃ Cd(PtO ₆)	280518	Hexagonal
Sr ₃ Zn(PtO ₆)	280519	Hexagonal
(Ba ₅ Br ₂)Ru ₂ O ₉	245668	Hexagonal
Ba ₄ (AgO ₂)(AuO ₄)	072329	Orthorhombic

Table: New materials confidently classified as superconducting (Stanev et. al, 2017)

Material	Predicted T_c (K)
CsBe(AsO ₄)	13.7
RbAsO ₂	8.0
KSbO ₂	10.2
RbSbO ₂	11.8
CsSbO ₂	10.1
AgCrO ₂	53.3
K _{0.8} (Li _{0.2} Sn _{0.76})O ₂	18.6
Cs(MoZn)(O ₃ F ₃)	20.5
Na ₃ Cd ₂ (IrO ₆)	17.4
Sr ₃ Cd(PtO ₆)	12.8
Sr ₃ Zn(PtO ₆)	12.4
(Ba ₅ Br ₂)Ru ₂ O ₉	17.0
Ba ₄ (AgO ₂)(AuO ₄)	56.7

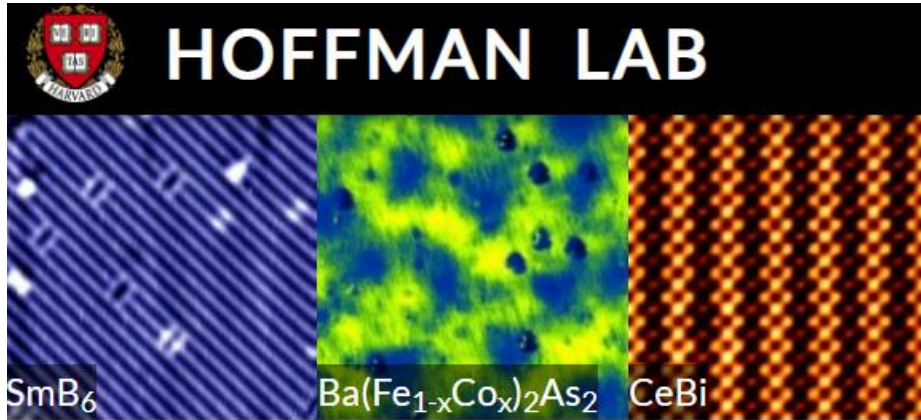
Table: T_c predictions for said list of potential superconductors (Hamidieh et. al, 2018)

Machine learning: Improvements

1. Approximate doped materials to parent compounds
2. What are more serviceable structural descriptors to use for ML?
3. Superconductivity depends sensitively on low-energy scale electronic state

Three take-aways

- 1) Finding a superconductor that functions at room-temperature would be *really awesome*
- 2) Approaches leveraging big data are promising and new
- 3) In exceptional materials discovery, computer-methods complement experiment nicely



Thank you!

Questions?